AMAN MASIPEDDI

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TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Dart, Kotlin, SQL, GRAPHQL gRPC, C++, HTML5
Frameworks/Libraries: Tensorflow, Scikit-Learn, LangChain, CrewAI, Pandas, Django, Flutter, Android, React, Node.js, CSS
Databases: PostgreSQL,DynamoDB, MongoDB, MySQL
Other Tools: AWS, AWS BedRock, AWS Sagemaker, AWS DynamoDB, Hugging Face, LangGraph, LangSmith, Jupiter Notebook, Docker, Kubernetes, XML, JSON, Kafka.

WORK EXPERIENCE

Happy Pillar — **New York City, US**
Senior Software Engineer — **Apr 2024 - Present**

- Serving as an early engineer at Happy Pillar(start-up) is a mental health care consumer app and platform that uses **NLP**, **Speech,** and **Sentiment Analysis** for **Behavioral insights** using **ML and AI** as per **HIPAA and CMS** compliance.
- Architected a network of Multi-AI Agents using **LangChain's LangGraph** Agentic Framework. These agents consist of general **NLP, Sentimental Analysis, Guard Rails,** and **Behavioral Summarization.**
- Achieved **99% availability** with **low latency** and **high throughput** by designing the **fault tolerance** structure for **AI Agents** using **Fallback micro language models**.
- Built a monitoring tool using **LangChain's LangSmith** platform to keep track of costs, token usage, and Risk Mitigations, which helped in providing cost and usage metrics to make the **Agent swarm** more efficient by **35%**.
- Proposed a project to build a proprietary Sentimental Analysis Language Model by fine-tuning the **OpenAI's Open source Whisper FM** and various Sentimental Analysis Language Models from **Hugging Face,** which can increase the Summarization accuracy by **40%**. Developed a prototype for the proposal.
- Architected and implemented scalable microservices built using **Django (Python)**, to serve as an web server connecting the Agent Swarm and the various Frontend points. Used **Postgres** as Database. Leveraged **Kubernetes(K8s)** and container pods for orchestrating the microservices and hosting them on **EKS(AWS)** to achieve high availability and throughput.
- Implemented comprehensive testing infrastructure (Unit, Integration, E2E) and CI/CD pipelines, helping improve application reliability from **75%** to **98%** error-free usage and increasing the code coverage to **93%** for the webserver.
- Built a prototype of **MLOps pipeline** using **AWS SageMaker** to improve the deployment standards and E2E testing of the AI Agents.

American Exchange Group — **New York City, US**
Senior Software Engineer — **Apr 2021 - Apr, 2024**

- Served as an early engineer in developing a white-label connected devices SaaS platform, contributing to its growth from inception to 2M+ monthly active users as part of a 5+ person engineering team
- Architected and implemented scalable microservices architecture built using **Django (Python)**, leading the migration from on-premise systems to AWS cloud-native infrastructure **(AWS ECS+Docker+Fargate)**. Designed and optimized database solutions by migrating non-transactional data from **Postgres** to **DynamoDB**, resulting in **30%** reduction in hosting/ management costs.
- Developed a personalized recommendation system using **KNN** and **Matrix Factorization** using **Pandas, Scikit-Learn, TensorFlow+Keras** on **AWS SageMaker**, incorporating user behavior patterns and historical data to suggest workout routines, leading to **40%** increase in product engagement.
- Engineered an **AI-powered virtual assistant** powered by a **semantic RAG engine** and built it by orchestrating a multi-agent system using the **LangGraph Agentic AI** framework and inferencing **LLAMA 3.1 FM** hosted on **AWS BedRock**.
- Implemented comprehensive testing infrastructure (Unit, Integration, E2E) and CI/CD pipelines, helping improve application reliability from 60% to 97% error-free usage and increasing the code coverage to 93%.

Oklahoma State University — **Oklahoma, US**
Software Developer — **Aug 2019 - Apr 2021**

- Engineered cross-platform mobile applications using **React Native** and **TypeScript**, delivering unified Android and iOS solutions that reduced development time and resources by 30%.
- Developed and deployed scalable microservices using **FastAPI (Python)** on **AWS EKS(K8s orchestration)** with containerized **Docker** architecture.
- Implemented **GraphQL API** integration using **AWS AppSync** with **DynamoDB** as the underlying database, enabling efficient real-time data synchronization.
- Established comprehensive automated testing infrastructure, including Unit and Integration tests with **Jest**, achieving **90%+** test coverage.

Free Lancer — **Hyderabad, India**
Android Developer — **July 2017 - March 2019**

- Leveraged **Google AdMob** in the mobile applications to increase user engagement and revenue.
- Designed and developed a feature-rich **Android** application using **Kotli,n** enabling users to participate in **Cricket Fantasy**, later adapted for **iOS** using **Swift**.
- Followed MVVM mobile architecture with **TTD** development using **Espresso** framework, which resulted in **90%** code coverage.
- Created a lightweight but effective **Pub/Sub** using **Websockets** over **Redis** to keep track of Fantasy leaderboards in real time.

EDUCATION

OKLAHOMA STATE UNIVERSITY — **Oklahoma, US**
Master of Science in Computer Science, GPA: 3.5/4.0 — **Apr 2021**

GANDHI INSTITUTE OF TECHNOLOGY AND MANAGEMENT — **Hyderabad, INDIA**
Bachelor of Technology in Computer Science and Engineering, GPA:3.65/4.0 — **May 2019**

CERTIFICATIONS

AWS Certified AI Practitioner(Early Adopter)

- Demonstrates knowledge of AI, machine learning (ML), and generative AI concepts and use cases
- https://www.credly.com/badges/143df321-4b56-4f92-8ffd-da658db87adc/linked_in_profile

PROJECTS

Autonomous UAVs

- This is my individual Creative Component project for my MS in computer science graduation. The main goal of this project is to automate the travel of a UAV(DJI drone) to its destination with the pilot. Achieved this by implementing **SLAM(Simultaneous Localization and Mapping)** Algorithm over the top **DJI UAV Protocol**.

Image Detection using ML (iOS Application)

- Developed an **iOS based Mobile Application** that can detect the type of image uploaded by the farmer to detect the crop affectiveness. Built in collaboration with Dept of Agriculture at OSU. Built the detection model by self hosting a Human-in-the-loop trained classification model.